

Intrusion Detection System (IDS) in IoT

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Intrusion Detection System (IDS) in IoT

- **Definition of IDS**

- An Intrusion Detection System (IDS) is a security mechanism that monitors network or system activities to detect unauthorized access and malicious behavior.
- It analyzes traffic patterns and generates alerts when suspicious activity is identified.

- **Why IoT Needs IDS**

- IoT devices are often resource-constrained and lack strong built-in security mechanisms.
- Therefore, IDS is essential to monitor communication and detect attacks in real time.

Types of Intrusion Detection Systems

- **Based on Deployment**
 - **Host-Based IDS (HIDS):**
 - Monitors activities of an individual device such as system logs and file integrity.
 - **Network-Based IDS (NIDS):**
 - Monitors network traffic across multiple devices.
- **Based on Detection Technique**
 - **Signature-Based IDS:**
 - Detects known attack patterns using predefined signatures.
 - **Anomaly-Based IDS:**
 - Detects deviations from normal behavior.
 - **Hybrid IDS:**
 - Combines both signature and anomaly detection techniques.

IoT Architecture with IDS Based on OSI Model and Attacks

- **OSI Layer–Wise Attacks and IDS Placement**
 - **Physical Layer**
 - Attacks: Device tampering, hardware manipulation
 - IDS Role: Detect abnormal device behavior or unauthorized access
 - **Data Link Layer**
 - Attacks: MAC spoofing, frame injection
 - IDS Role: Monitor unusual MAC address changes and abnormal traffic patterns
 - **Network Layer**
 - Attacks: IP spoofing, routing attacks, Denial-of-Service (DoS)
 - IDS Role: Monitor packet flow, IP anomalies, traffic spikes
 - Common Placement: Gateway (Raspberry Pi)
 - **Transport Layer**
 - Attacks: Port scanning, SYN flood attacks
 - IDS Role: Analyze TCP/UDP behavior and connection attempts
 - **Application Layer**
 - Attacks: Data injection, malware, unauthorized API access
 - IDS Role: Inspect payload patterns and application-level anomalies

Study of IoT Using Raspberry Pi and ESP32-C5

- **Hardware Role in IoT Architecture**
 - **ESP32-C5 (Sensor Node)**
 - Low-power microcontroller with Wi-Fi capability
 - Collects environmental or sensor data
 - Transmits data to gateway (Raspberry Pi)
 - Limited memory and processing power
 - Reason: Suitable for distributed IoT deployment.
 - **Raspberry Pi (Gateway)**
 - Higher computational capability
 - Connects sensor nodes to internet/cloud
 - Runs IDS monitoring software
 - Performs traffic analysis and alert generation

Machine Learning Used in IDS for IoT

- Machine Learning enhances IDS by enabling automatic detection of unknown and evolving attacks.
- It identifies abnormal traffic patterns by learning from network data.
- **ML in IoT Architecture**
 - ML models can be deployed at the edge (Raspberry Pi) or cloud for real-time traffic classification.
 - Lightweight algorithms are preferred due to limited computational resources in IoT systems.
- **Suitable Lightweight Algorithms**
 - Decision Tree, K-Nearest Neighbors (KNN), and optimized Random Forest models provide efficient detection with low overhead.

*Thank
you!*